ECE 444 Equation Sheet

Antennas, Phased Arrays, and Radar Systems | Department of Electrical and Computer Engineering, United States Air Force Academy

Module 1 Foundations

Waves and power

$$k = \frac{2\pi}{\lambda}$$

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx 377\ \Omega$$

$$S = \frac{|\mathbf{E}|^2}{2\eta_0}$$

$$U(\theta,\phi) = r^2 S$$

$$P_{\text{rad}} = \oint U(\theta,\phi)\, d\Omega$$

Directivity and gain

$$D = \frac{4\pi U_{\max}}{P_{\text{rad}}}$$

$$D \approx \frac{41{,}253}{\theta_E^\circ\, \theta_H^\circ}$$

$$\Omega_A = \frac{P_{\text{rad}}}{U_{\max}}$$

$$D = \frac{4\pi}{\Omega_A}$$

$$\eta_{\text{rad}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}$$

$$G = \eta_{\text{rad}}\, D$$

$$A_e = \frac{\lambda^2 G}{4\pi}$$

Polarization

$$\text{AR} = \frac{E_{\text{major}}}{E_{\text{minor}}}$$

$$\text{PLF} = |\hat{\boldsymbol{\rho}}_w \cdot \hat{\boldsymbol{\rho}}_a|^2$$

$$\text{PLF} = \cos^2\psi \quad \text{(two linear, tilt } \psi\text{)}$$

Module 1 Foundations, cont.

Impedance and matching

$$Z_{\text{in}} = R_{\text{rad}} + R_{\text{loss}} + jX_{\text{in}}$$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\text{VSWR} = \frac{1+|\Gamma|}{1-|\Gamma|}$$

$$\text{ML} = -10\log_{10}\left(1 - |\Gamma|^2\right)$$

$$Z_1 = \sqrt{Z_0 Z_L}$$

$$Q = \sqrt{\frac{R_{\text{high}}}{R_{\text{low}}} - 1}$$

Field regions

$$r = \frac{\lambda}{2\pi}$$

$$r < 0.62\sqrt{\frac{D^3}{\lambda}}$$

$$r \geqslant \frac{2D^2}{\lambda}$$

Radiation integral

$$\mathbf{A} = \frac{\mu}{4\pi}\int \mathbf{J}\, \frac{e^{-jkR}}{R}\, dV'$$

$$R \approx r - \hat{\boldsymbol{r}} \cdot \mathbf{r}'$$

$$\mathbf{N} = \int \mathbf{J}\, e^{jk\hat{\boldsymbol{r}}\cdot\mathbf{r}'}\, dV'$$

$$E_\theta = -j\omega\mu\, \frac{e^{-jkr}}{4\pi r}\, N_\theta$$

Link budget

$$\text{EIRP} = P_t G_t$$

$$\text{FSPL} = 20\log_{10}\left(\frac{4\pi R}{\lambda}\right)$$

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2$$

Module 2 Antenna Types

Dipoles and monopoles

$$R_{\text{rad}} = 80\pi^2\left(\frac{\ell}{\lambda}\right)^2 \quad \text{(uniform } I\text{)}$$

$$R_{\text{rad}} = 20\pi^2\left(\frac{\ell}{\lambda}\right)^2 \quad \text{(triangular } I\text{)}$$

$$Z_{\text{in}} = 73 + j42.5\ \Omega$$

$$\ell_{\text{res}} \approx 0.48\lambda$$

$$Z_{\text{mono}} = \tfrac{1}{2} Z_{\text{dipole}} \qquad D_{\text{mono}} = 2D_{\text{dipole}}$$

Canonical patterns

Antenna	$\lvert \boldsymbol{F(\theta)} \rvert$	$\boldsymbol{D}$	$\boldsymbol{\theta}_{\text{HP}}$
Isotropic	1	0 dBi	—
Short dipole, loop	$\sin\theta$	1.76 dBi	90°
$\lambda/2$ dipole	$\cos\left(\frac{\pi}{2}\cos\theta\right)/\sin\theta$	2.15 dBi	78°
$\lambda/4$ monopole	same, upper half	5.15 dBi	39°

Loops

$$R_{\text{rad}} = 20\pi^2\left(\frac{C}{\lambda}\right)^4$$

$$Q_{\min} = \frac{1}{(ka)^3} + \frac{1}{ka}$$

Slots (Babinet)

$$Z_{\text{slot}} Z_{\text{dipole}} = \frac{\eta_0^2}{4} = 3.55 \times 10^4\ \Omega^2$$

$$Z_{\text{slot}} \approx 485\ \Omega \quad (\lambda/2 \text{ slot})$$

Patches and apertures

$$L \approx \frac{\lambda_0}{2\sqrt{\epsilon_{\text{eff}}}}$$

$$G = \eta_{\text{ap}}\, \frac{4\pi A}{\lambda^2}$$

$$\tan\left(\frac{\theta_0}{2}\right) = \frac{1}{4(f/D)}$$

$$\Delta G = -685.8\left(\frac{\sigma}{\lambda}\right)^2 \text{dB}$$

Measurement

$$G_{\text{AUT}} = G_{\text{std}} + (P_{\text{AUT}} - P_{\text{std}})$$

Module 3 Arrays and Beamforming

Array factor

$$\mathrm{AF}(\psi) = \sum_{n=0}^{N-1} a_n e^{jn\psi}$$

$$\psi = kd\sin\theta + \beta$$

$$|\mathrm{AF}| = \frac{\sin(N\psi/2)}{N\sin(\psi/2)}$$

$$F(\theta) = \mathrm{EF}(\theta)\cdot\mathrm{AF}(\theta)$$

Beam steering

$$\beta = -kd\sin\theta_0$$

$$\Delta\phi = 360^\circ\left(\frac{d}{\lambda}\right)\sin\theta_0$$

$$\beta = 0 \text{ (broadside)} \qquad \beta = -kd \text{ (endfire)}$$

Beamwidth and directivity

$$\sin\theta_n = \frac{n\lambda}{Nd}$$

$$\theta_{\mathrm{HP}} = \frac{0.886\,\lambda}{Nd}$$

$$\theta_{\mathrm{HP}}(\theta_0) = \frac{\theta_{\mathrm{HP}}(0)}{\cos\theta_0}$$

$$D = \frac{2Nd}{\lambda}$$

Gain and scan loss

$$G(\theta_0) = G_e(\theta_0) + 10\log_{10} N$$

$$G_e(\theta_0) = G_e(0) + 10\log_{10}(\cos\theta_0)$$

Module 3 Arrays, cont.

Amplitude tapering

$$\eta_t = \frac{(\sum a_n)^2}{N\sum a_n^2}$$

$$\Delta P_{\mathrm{peak}} = 20\log_{10}\left(\frac{\sum a_n}{N}\right)$$

Taper	SLL	θ_{HP}	η_t
Uniform	−13.3 dB	13.2°	1.00
Taylor $\bar{n}$=4	−28.3 dB	16.8°	0.85
Hann	−31.8 dB	19.1°	0.75
Blackman	−50.5 dB	21.7°	0.66
Chebyshev	by design	22.9°	0.62

Grating lobes

$$\sin\theta_g = \sin\theta_0 \pm \frac{m\lambda}{d}$$

$$d < \frac{\lambda}{1+|\sin\theta_0|}$$

Squint and quantization

$$\Delta\theta \approx -\frac{\Delta f}{f}\tan\theta_0$$

$$\mathrm{LSB} = \frac{360^\circ}{2^B}$$

$$\mathrm{SLL}_q \approx -6B \text{ dB}$$

Null steering

$$\mathbf{w} = \mathbf{w}_d - r_n\mathbf{w}_i$$

$$r_n = \frac{\mathbf{w}_i^H\mathbf{w}_d}{\mathbf{w}_i^H\mathbf{w}_i}$$

$$\mathbf{w}_{\mathrm{MVDR}} = \frac{\mathbf{R}^{-1}\mathbf{s}}{\mathbf{s}^H\mathbf{R}^{-1}\mathbf{s}}$$

Reference

Constants

$c = 2.998\times10^8$ m/s

$\eta_0 = 376.7\ \Omega$

$\epsilon_0 = 8.854\times10^{-12}$ F/m $\qquad \mu_0 = 4\pi\times10^{-7}$ H/m

4π sr = 41,253 deg^2

Decibels

$X_{\mathrm{dB}} = 10\log_{10} X$ (power)

$X_{\mathrm{dB}} = 20\log_{10}|X|$ (field)

dBi = dBd + 2.15

3 dB = ×2 $\qquad$ 10 dB = ×10

6 dB = ×4 $\qquad$ 20 dB = ×100

VSWR, reflection, and loss

VSWR	$\lvert\Gamma\rvert$	RL	ML
1.0	0	∞	0
1.5	0.20	14.0 dB	0.18 dB
2.0	0.33	9.5 dB	0.51 dB
3.0	0.50	6.0 dB	1.25 dB

Wavelength

f	λ	Where
300 MHz	1.00 m	scaling anchor
915 MHz	32.8 cm	L7 dipole
2.45 GHz	12.2 cm	L10 patch
10.25 GHz	2.93 cm	PHASER

Typical gains (sanity check)

Antenna	G
Isotropic	0 dBi
$\lambda/2$ dipole	2.15 dBi
Patch	6–8 dBi
Standard-gain horn	15–20 dBi
1 m dish, 10 GHz	≈ 38 dBi

Course array (ADALM-PHASER)

$N = 8$ elements

$d = 14$ mm $\approx \lambda/2 \qquad f = 10.25$ GHz